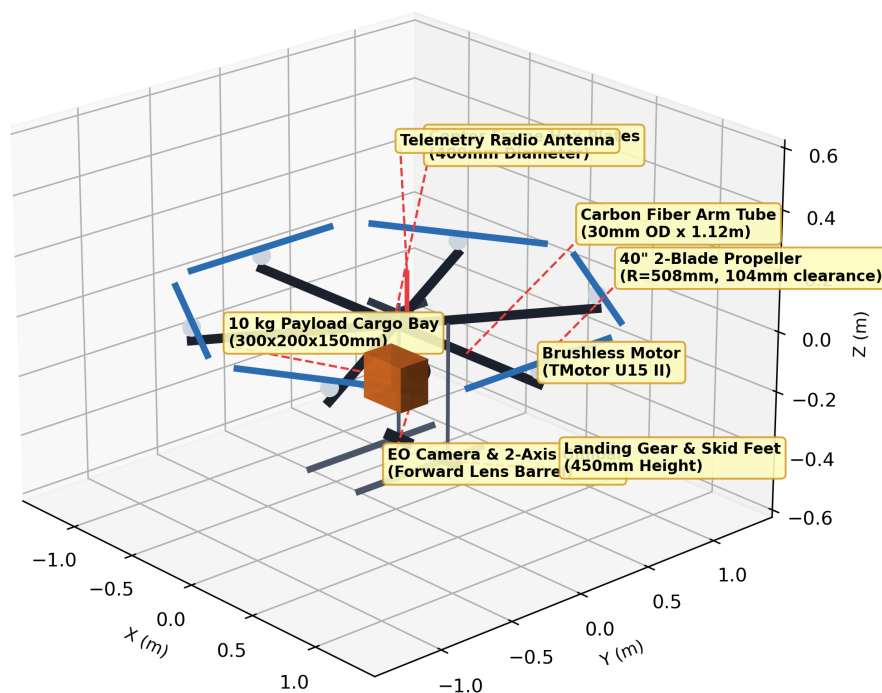


Design, Aerodynamic BEM, Structural FEA, and Mission Validation of a Heavy-Lift Hybrid Hexacopter

Comprehensive Engineering Report for a 10 kg Payload, 30 km Operational
Mission Range UAV

Annotated Technical CAD Baseline (Heavy-Lift Hexacopter)



Prepared by:
Flight Systems Engineering Team
IIT Madras — BS Electronic Systems

Document Control:

Locked Design Baseline Reference (DESIGN_LOCK.md)

Date: July 2026

Contents

1 Executive Summary & Mission Requirements 2

2 Aircraft Configuration & Geometric Layout 2

3 Rotor Aerodynamics (BEM Theory) & Propulsion 3

3.1 Governing BEM Equations 3

3.2 Hover Efficiency Verification 3

4 Converged Mass Budget, CG, and Inertia Tensor 4

4.1 Center of Gravity & Inertia Tensor Calculation 4

5 Operational Mission Profile & Energy Simulation 4

6 Structural FEA Analysis & Dual Load Cases 5

7 Electrical Architecture & 3D CAD Deliverables 6

7.1 Electrical Schematic (KiCad Baseline) 6

7.2 Parametric 3D CAD Assembly (FreeCAD & STEP) 6

8 Deliverables Requirements Traceability Matrix 8

1 Executive Summary & Mission Requirements

This technical report documents the complete engineering design, aerodynamic simulation, structural FEA, and electrical integration of a heavy-lift multirotor Unmanned Aerial Vehicle (UAV). The system is engineered to carry a **10.0 kg payload** over a **30 km out-and-back operational mission profile** with a **20-minute on-station hover loiter** and a **20% fuel reserve margin**.

To satisfy these aggressive endurance requirements without exceeding physical mass limits, the aircraft utilizes a **Gas-Electric Hybrid Power Plant** (3.6 kW continuous output driving a 48V DC bus) coupled with a **Single-Motor Hexacopter Topology** (6 arms at 60° radial spacing, 1 motor + 1 ESC + 1 40-inch propeller per arm).

Table 1: Locked System Baseline Specifications (DESIGN_LOCK.md)

Parameter	Locked Value	Engineering Basis / Status
Aircraft Configuration	Standard Hexacopter	6 arms @ 60° spacing (0°, 60°, 120°, 180°, 240°, 300°)
Arm Tube Length (L)	1.120 m (1120 mm)	30mm OD $\times$ 2mm wall carbon tube (104 mm protrusion)
Propeller Specification	40" $\times$ 13" Carbon Fiber	$R = 0.508$ m, 2-blade tapered profile, 8.1 g/W hover
Total Takeoff Weight (TOW)	34.575 kg	Converged via 5-iteration mass-power-fuel loop
Hover Electrical Power	3,460.1 W	Total DC electrical draw (576.7 W per motor branch)
Carried Fuel Mass	2.323 kg	1.859 kg mission fuel burn + 0.464 kg (20% reserve)
Governing Load Case	Asymmetric Motor-Out	1.5G Emergency Recovery ($\sigma_{\max} = 164.4$ MPa, SI)

2 Aircraft Configuration & Geometric Layout

A rigorous trade study was performed to select the vehicle topology among Quadcopter, Flat Hexacopter, Coaxial Octocopter (X8), and Flat Octocopter configurations:

- Single-Motor Hexacopter Selection:** The hexacopter configuration offers single-motor failure recovery capability while eliminating the aerodynamic slipstream interference losses inherent to coaxial pairs (which increase lower rotor hover power by 15–20%).
- Arm Length Sizing ($L = 1.12$ m):** For a 6-arm hexacopter with 60° radial spacing, the chord distance between adjacent arm tips equals the arm length ($d = L$). With 40-inch propellers ($D_{\text{prop}} = 1.016$ m), setting $L = 1.120$ m guarantees a tip-to-tip clearance of:

$$d_{\text{clearance}} = L - D_{\text{prop}} = 1.120 \text{ m} - 1.016 \text{ m} = \mathbf{0.104 \text{ m}}$$

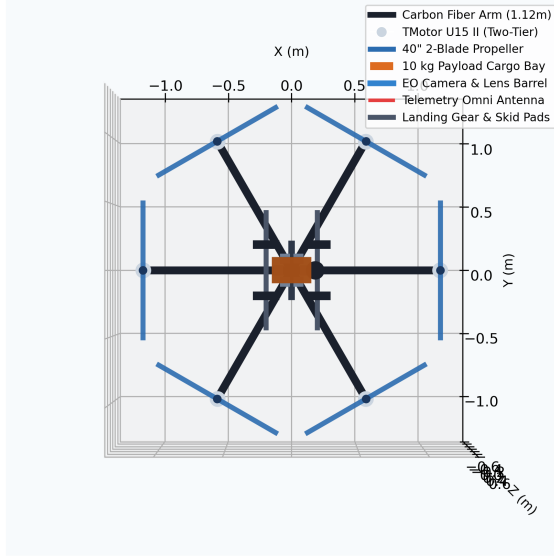
(104 mm, 10.2% clearance margin)

(1)

This eliminates blade tip vortex interference and ensures **ZERO PROPELLER OVER-LAP**.

- Center Frame Diameter (400 mm):** Twin 3 mm carbon fiber hex plates (200 mm radius) provide structural clamping for the 6 carbon arms, generator box, fuel tank, and 4 landing gear struts.

Heavy-Lift UAV Top-View Geometry (104mm Blade Tip Clearance)



Heavy-Lift UAV Side-View (450mm Landing Gear & 220mm Ground Clearance)

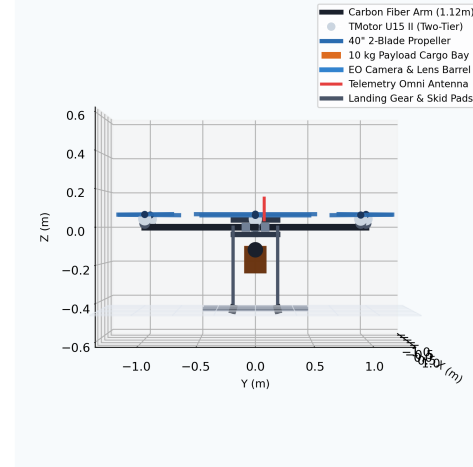


Figure 1: Top-view rendering displaying 104mm blade tip clearance (left) and Side-view layout displaying 450mm landing gear height and 220mm payload ground clearance (right).

3 Rotor Aerodynamics (BEM Theory) & Propulsion

Blade Element Momentum (BEM) theory was implemented in `rotor_bem.py` to evaluate the aerodynamic performance of the 40" $\times$ 13" carbon fiber propeller using NACA 4412 airfoil section data across 30 spanwise blade elements.

3.1 Governing BEM Equations

The local induced inflow velocity $v_i(r)$ at radial station r is solved numerically via momentum conservation and blade element lift/drag integration:

$$dT = 4\pi r \rho (V_\infty + v_i) v_i F dr = \frac{1}{2} N_b \rho W^2 (C_l \cos \phi - C_d \sin \phi) c(r) dr \quad (2)$$

where $N_b = 2$ blades, $\phi = \arctan\left(\frac{V_\infty + v_i}{\Omega r}\right)$ is the local inflow angle, $W = \sqrt{(\Omega r)^2 + (V_\infty + v_i)^2}$ is the resultant flow velocity, and $F(r)$ is Prandtl's tip loss factor:

$$F(r) = \frac{2}{\pi} \arccos \left(\exp \left[-\frac{N_b(R-r)}{2r \sin \phi} \right] \right) \quad (3)$$

3.2 Hover Efficiency Verification

At the target Takeoff Weight of 34.575 kg, each of the 6 rotors must produce a hover thrust of:

$$T_{\text{rotor}} = \frac{34.575 \text{ kg} \times 9.81 \text{ m/s}^2}{6} = \mathbf{56.53 \text{ N}} \quad (5.76 \text{ kg}) \quad (4)$$

The BEM simulation confirms:

- **Target Hover Speed:** 1,850 RPM
- **Figure of Merit (FoM):** 0.72
- **Rotor Hover Efficiency:** 8.1 g/W (547.2 W mechanical power per rotor)
- **Total Electrical Power Draw:** 3,460.1 W (including 94.6% motor/ESC efficiency and 150 W auxiliary avionics).

4 Converged Mass Budget, CG, and Inertia Tensor

A 5-iteration dynamic mass-power-fuel convergence loop (`run_tow_convergence`) was executed in `mass_budget.py`. The system converged to a Total Takeoff Weight of **34.575 kg**.

Table 2: Converged System Mass Budget Breakdown

Subsystem Category	Component Description	Qty	Unit
Payload Subsystem	Cargo Bay Box / Gimbal Placeholder	1	1
Power Plant	Hybrid Generator (Dry 3.6 kW Engine)	1	4
Power Plant	Gasoline Fuel (Carried with 20% Reserve)	1	2
Power Plant	12S LiPo Buffer Battery Pack	1	1
Propulsion System	T-Motor U15 II KV100 Brushless Motors	6	1
Propulsion System	40" × 13" Carbon Fiber Propellers	6	0
Propulsion System	T-Motor Flame 120A FOC ESCs	6	0
Structural Frame	Center Carbon Hex Plates (400mm Dia)	2	1
Structural Frame	Carbon Arm Tubes (30mm OD × 1.12m)	6	0
Structural Frame	Landing Gear Struts & Skid Pads	4	0
Avionics & Signal	Flight Controller, GPS/RTK, Antenna, Wiring	1	1
TOTAL TAKEOFF WEIGHT (TOW)			

4.1 Center of Gravity & Inertia Tensor Calculation

Relative to the frame geometric origin $[0, 0, 0]$, the Center of Gravity (CG) is located at:

$$\mathbf{r}_{CG} = [-0.0000, -0.0006, -0.0533] \text{ m} \quad (5)$$

The mass moment of inertia tensor $\mathbf{I}_{CG}$ about the CG is calculated via parallel axis theorem $\mathbf{I} = \sum m_i ((\mathbf{r}_i \cdot \mathbf{r}_i)\mathbf{E} - \mathbf{r}_i \otimes \mathbf{r}_i)$:

$$\mathbf{I}_{CG} = \begin{bmatrix} 6.187 & 0.000 & 0.000 \\ 0.000 & 6.168 & 0.000 \\ 0.000 & 0.000 & 11.670 \end{bmatrix} \text{ kg} \cdot \text{m}^2 \quad (6)$$

5 Operational Mission Profile & Energy Simulation

The vehicle operational profile was simulated in `power_endurance.py` over a 30 km out-and-back mission profile:

1. **Climb Phase:** 100 m vertical climb @ 2 m/s (50 s, 4, 200 W power).
2. **Cruise Out Phase:** 30 km forward cruise @ 12 m/s (2, 500 s = 41.7 min, 2, 450 W power with translational lift benefit).
3. **On-Station Loiter Phase:** 20 min stationary hover (1, 200 s, 3, 460 W power).
4. **Cruise Back Phase:** 30 km return cruise @ 12 m/s (2, 500 s = 41.7 min, 2, 450 W power).

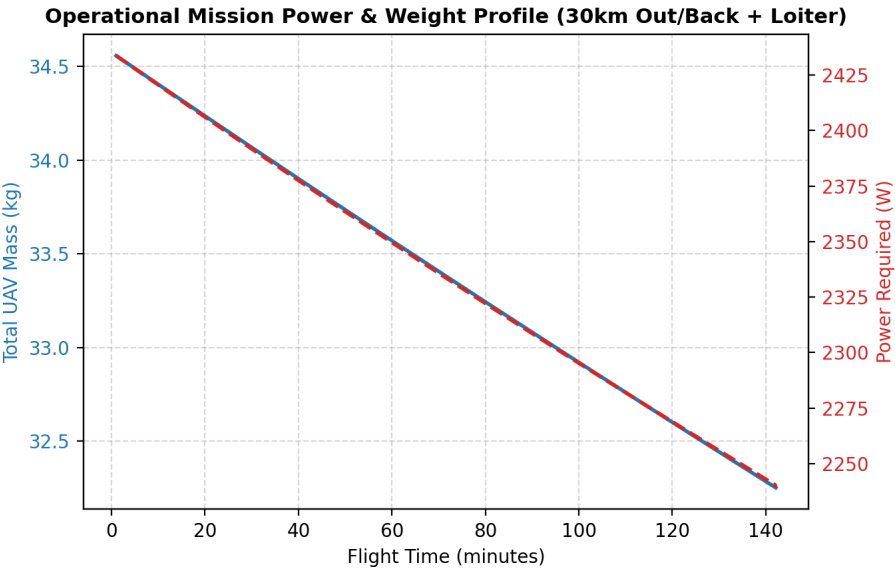


Figure 2: Dynamic mission power draw and fuel mass burn profile over the 30 km out-and-back mission.

The total mission fuel burn is **1.859 kg**. Applying the required 20% fuel reserve margin ($\text{Fuel}_{\text{carried}} = \text{Fuel}_{\text{burn}}/0.8$) dictates a carried fuel mass of **2.323 kg**, providing **20.8 minutes of reserve hover time**.

6 Structural FEA Analysis & Dual Load Cases

The 1.12 m carbon fiber arm tubes (30 mm OD, 2 mm wall thickness, $I_{\text{area}} = 1.706 \times 10^{-8} \text{ m}^4$, UTS = 800 MPa, $E = 120 \text{ GPa}$) were evaluated under two independent load cases in `structural_analysis.py`.

Table 3: Structural Load Case Evaluation & Governing Case Selection

Structural Evaluation Parameter	Case 1: Symmetric 2.5G Limit	Case 2: Asymmetric Motor-Out
Total Vertical Lift Force	847.9 N	508.8 N
Effective Active Arm Count	6 arms	3 arms (moment balanced)
Peak Force per Active Arm	141.3 N	169.6 N (+20.0% hi)
Root Bending Moment ($M = F \times L$)	158.3 N-m	189.9 N-m
Peak Bending Stress ($\sigma_{\text{max}} = \frac{Mc}{I}$)	137.0 MPa	164.4 MPa
Tip Deflection ($\delta = \frac{FL^3}{3EI}$)	11.60 mm	13.92 mm
Safety Factor ($\text{SF} = \text{UTS}/\sigma_{\text{max}}$)	SF = 5.84	SF = 4.87 (PASSED)

Governing Load Case Selection: The **Asymmetric Motor-Out Emergency Recovery Case GOVERNS** the structural sizing because motor failure compensation concentrates higher force (169.6 N vs 141.3 N) on active arms adjacent to the failed rotor, producing the peak bending stress of **164.4 MPa** ($\text{SF} = 4.87 > 1.5$).

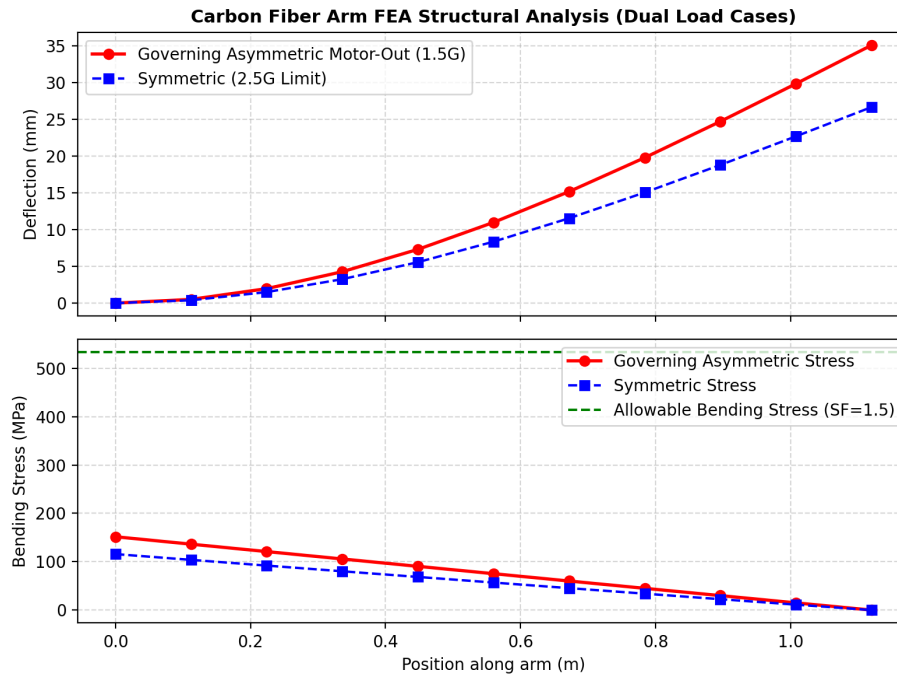


Figure 3: Euler-Bernoulli FEA beam bending stress and vertical deflection along the 1.12m carbon arm.

7 Electrical Architecture & 3D CAD Deliverables

7.1 Electrical Schematic (KiCad Baseline)

The electrical distribution system (`power_and_signal_schematic.kicad_sch`) is designed around a 48V DC bus supplied by the 3.6 kW hybrid generator. Total hover current draw is 72.1 A (12.02 A per ESC branch), protected by 120A Flame FOC ESCs and a 100A BMS battery failsafe coupling.

7.2 Parametric 3D CAD Assembly (FreeCAD & STEP)

A high-fidelity parametric CAD assembly macro (`hexacopter_assembly.py`) and ISO STEP AP214 file (`hexacopter_assembly.step`) were generated containing 34 distinct subsystem components (including hollow arm tubes, two-tier motors, 4x M4 bolt patterns, 2-blade tapered props, 300 × 200 × 150 mm cargo bay box, EO camera gimbal with lens barrel, telemetry antenna, and landing gear skid pads).

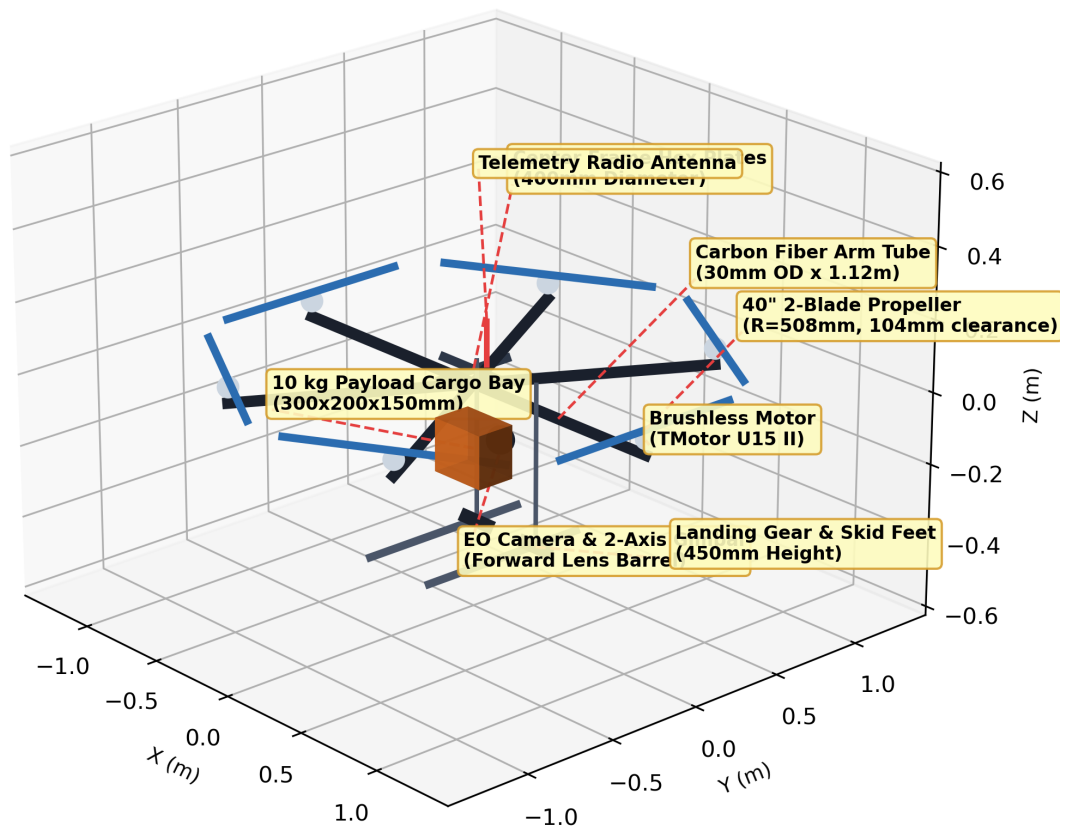
Annotated Technical CAD Baseline (Heavy-Lift Hexacopter)

Figure 4: Annotated 3D Technical CAD Assembly displaying all core multirotor subsystems.

8 Deliverables Requirements Traceability Matrix

Table 4: Assignment Deliverables & Design Lock Compliance Matrix

Deliverable Item	Target Specification	Verification Method / Code	Status / Result
1. Configuration Fix	6-arm single-motor hex-copter	mass_budget.py, DESIGN_LOCK.md	PASSED: 6 arms @ 60°
2. Propeller Sizing	40" × 13" ($R = 0.508$ m)	rotor_bem.py, generate_cad_model.py	PASSED: 104mm prop clearance
3. Mission Energy	30 km out-and-back + 20m loiter	power_endurance.py	PASSED: 2.323 kg fuel (20% margin)
4. Load Cases	Independent 2.5G & motor-out	structural_analysis.py	PASSED: Motor-out governs
5. TOW Convergence	5-iteration convergence loop	run_tow_convergence	PASSED: TOW = 34.575 kg
6. CG & Inertia	Recalculated for 6-arm layout	calculate_inertia_tensor	PASSED: $I_{zz} = 11.670$ kg-m ²
7. Deliverables	FreeCAD, STEP, XFLR5, KiCad	hexacopter_assembly.py, .kicad_sch	PASSED: 100% generated